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Topic :Facial recognition sysyem

FACIAL RECOGNITON SYSTEM

TAB 1: Introduction

Facial Recognition is a biometric technology.
It identifies or verifies a person using facial features.
Widely used in security, attendance systems, and mobile authentication.

TAB 2 : Objectives

Detect human faces from images or video.
Recognize and identify individuals.
Improve security and automation.
Maintain attendance records automatically.

TAB 3 : Problem Statement

Manual identification is time-consuming.
Traditional attendance systems are inefficient.
Need for secure and automated authentication.

TAB 4: System Architecture

Image Capture
Face Detection
Feature Extraction
Face Matching
Result Display

TAB 5: Technologies Used

Python
OpenCV
NumPy
Face Recognition Library
SQLite Database

TAB 6: Methodology

Capture facial images.
Train the system with known faces.
Detect faces in real-time.
Compare with stored database.
Display identified person's details.

TAB 7: Implementation

```
import cv2

img = cv2.imread("image.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

face_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades +
    'haarcascade_frontalface_default.xml'
)

faces = face_cascade.detectMultiScale(gray, 1.3, 5)

for (x,y,w,h) in faces:
    cv2.rectangle(img,(x,y),(x+w,y+h),(255,0,0),2)

cv2.imshow("Face Detection", img)
cv2.waitKey(0)
Python
```

TAB 8: Advantages

Fast and accurate identification.
Improved security.
Contactless authentication.
Automated attendance management.

TAB 9: Applications

Attendance Systems
Mobile Phone Unlocking
Airport Security
Banking Authentication
Smart Surveillance

TAB 10: Future Enhancements

AI-based recognition.
Masked face detection.
Cloud database integration.
Multi-user real-time recognition.

TAB 11: Conclusion

Facial Recognition provides secure and efficient identification.
Reduces manual work and improves accuracy.
Useful in various real-world applications.

THANK YOU ☺